REVIEW OF APU COLD SOAKED ISSUE FOR ET OPERATIONS

Based on the data from Boeing and Honeywell, currently only few airports on our operations are affected by APU cold soaked issue . Reducing the routes will help us to accurately monitor the implementation. Additionally, it improves flight crew/technician workload and at the same time reducing unnecessary cost. To track the implementation an e-form is prepared to be filled by flight crew.

Boeing's and Honeywell's research data :
AHM data shows No-Start on ground with the following conditions:
-After landing on a long cold soaked flights (3.5+ hours)
-B737MAX has 2.4 times more failure rate than B737NG

 # Weather conditions for a failed APU:
Based on for approximately 400 departures and 400 arrivals:
-Temperatures 55°F to 65°F (13-18c), Majority 12-22c
-Humidity (>60%), Majority >70%
* The weather analysis also determined that ~93% of APU no start event arrivals encountered clouds and ~72% encountered rain.

Maintenance responsibility
#Monitor if the issue occurs more frequently on an **individual aircraft** with high APU hours/cycles. **Cold soaked starting issues do occur on APUs that are high time.**

#Before starting the APU when the airplane has been on the ground for sufficient time to accumulate snow or ice, **make sure the APU inlet door flap and the APU inlet are free of ice and snow before the APU is started.**

Specific routes after November
Note: Based on our maintenance analysis, the chosen airports are more susceptible to APU failure and have higher frequency than other routes, supporting Boeing and Honeywell's research data.

1.ADD-IST
2.ADD-ATH

FLIGHT CREW OPERATING PROCEDURE:
Note: Apply the following procedure only on routes to **IST** and **ATH** flights and fill out ‘APU cold soaked issue ‘ e-form.

1. Set the APU Master Switch to “ON” to open the APU inlet door with the SOP call, ” Cabin crew be seated for landing “

2. Wait at least 10-15 minutes between opening the APU inlet door and starting the APU.
Note: The intent is to bring APU internal temperatures above freezing.

3. When ready to start the APU, move the APU Master Switch from “ON” to “START” per normal operating procedures.

Risk Evaluation

1. APU being forgotten to be started after turned to ON position
2. Incase of GA and diversion, there is a fuel penalty of 2.4% for the open APU door

For a diversion airport of 1 hour this roughly equals to extra 100 kg